



Statistical Reasoning 5

PLAN FOR TODAY!

- ☐ Sampling, Designing Studies & Sampling Distributions
- ☐ Statistical Inference

Announcement

- The prompt for your semester project is online. Please familiarize yourself with the prompt.
 - **Deliverable 1:** March 24th, 11:59 pm
 - **Deliverable 2:** April 5th, 11:59 pm
 - **Deliverable 3:** April 19th, 11:59 pm
 - **Deliverable 4:** May 1st, 11:59 pm

Deliverable 2

- Please submit a document containing the following:
 - Introduction & Research Question (including Literature Review)
 - Hypothesis
- Please come discuss this deliverable with me before submission. This **meeting** is **mandatory**.

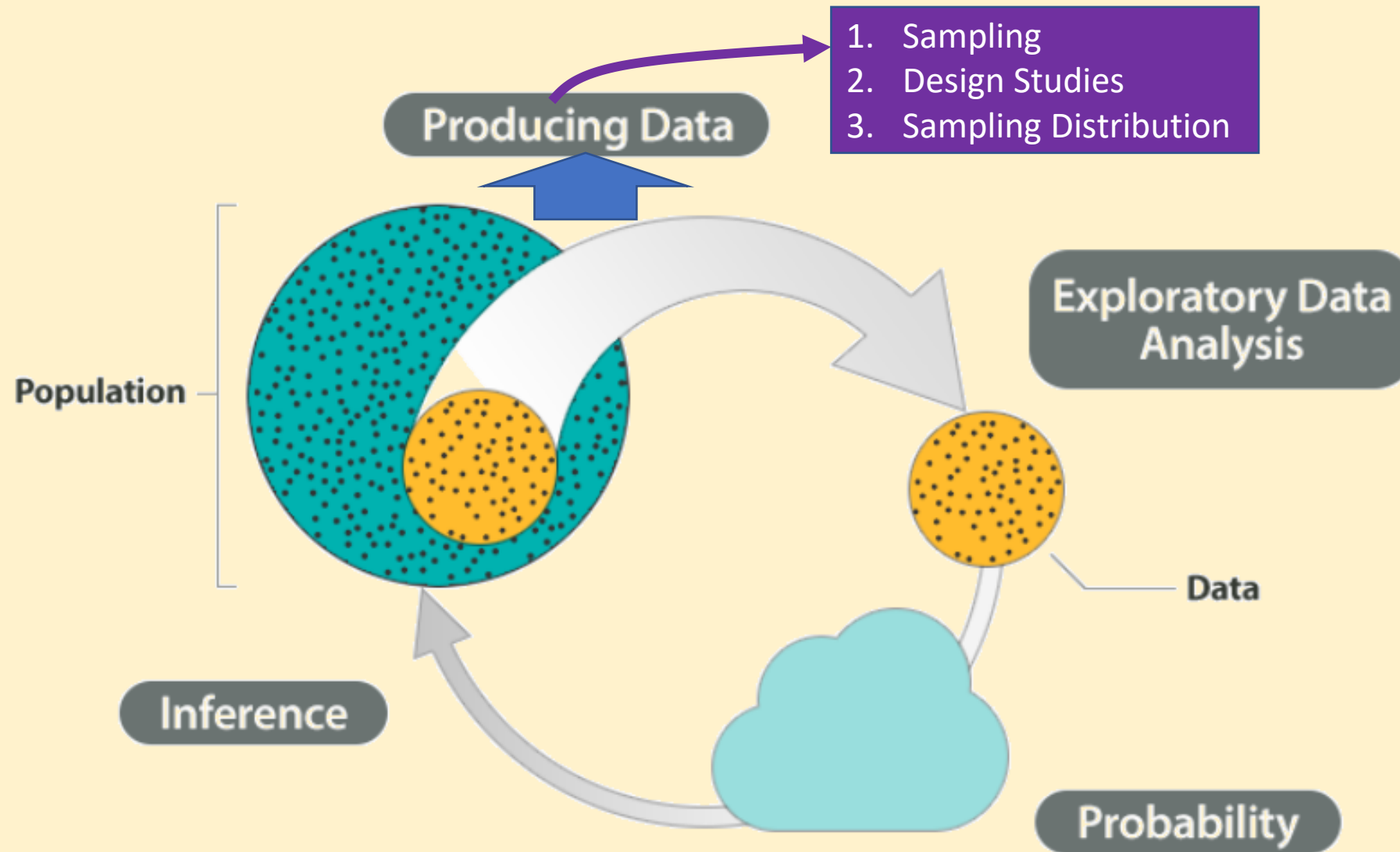
Rubric: Deliverable 2

Deliverable 2

Introduction and Refined Research Question				
Criteria	> 85% Exemplary	75% – 85% Proficient	65% – 75% Developing	< 65% Needs Improvement
Research Question & Hypothesis (50%)	Question is novel, focused, clearly quantitative, and testable. Hypothesis is specific, directly stems from the question, and makes a clear, directional prediction.	Question is focused and testable. Hypothesis is clear and testable but may lack specificity or a strong theoretical link.	Question is too broad, vague, or not fully quantitative. Hypothesis is unclear, not directly testable, or does not logically follow from the question.	Question is descriptive, not explanatory, or is untestable. Hypothesis is missing or completely unrelated to the question.
Structure, Clarity, and Writing (25%)	Exceptionally well-organized and a pleasure to read. Flawless grammar, spelling, and professional tone. Perfectly follows the required structure.	Well-organized and clear. Minor errors in grammar or structure do not impede understanding.	Disorganized or difficult to follow at times. Numerous errors in writing or structure detract from the content.	Poorly organized and unclear. Pervasive errors make the introduction difficult to understand.
Literature & Scholarly Context (25%)	5 or more relevant, high-quality scholarly sources are seamlessly integrated to justify the research question and hypothesis. References are flawless	3-5 relevant sources are used and cited correctly. The link between literature and the proposal is clear.	Sources are insufficient in number, not scholarly, or poorly related to the topic. Connection to the proposal is weak. Citations have errors.	Lacks scholarly sources or uses non-credible sources (e.g., blogs, wikis). Citations are missing or severely incorrect.

Sampling, Designing Studies & Sampling Distributions

Last time!



LONG
TIME
AGO

STATISTICS – THE BIG PICTURE

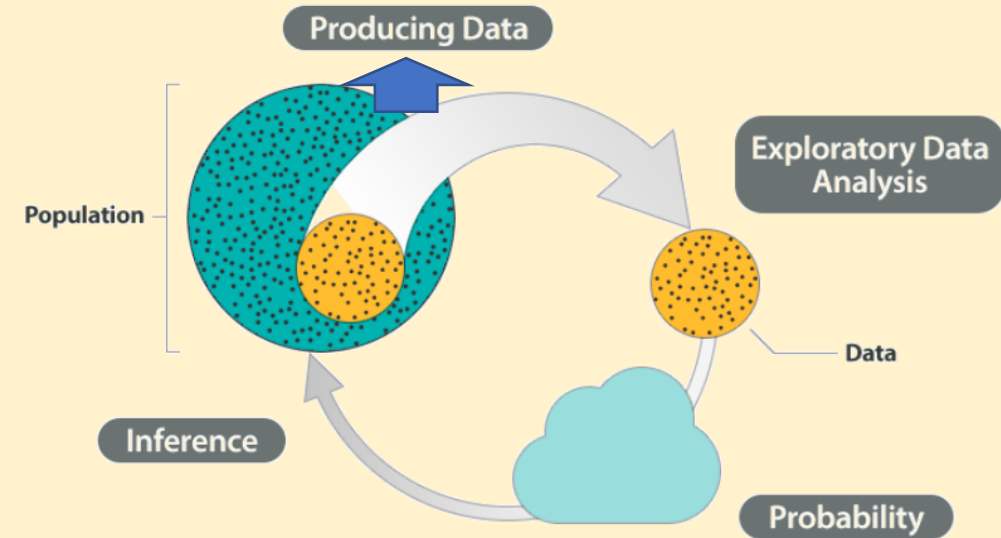
Last time!

- **Population (Target Group):** The whole collection of individuals under the study.
- **Sample (Sample Member):** The observed members of the target group.
 - Note, we should make an effort to choose a sample in such a way that it will represent the population well.
 - For example, if we choose a sample from the population of U.S. adults, and ask their opinion about the death penalty, we do not want our sample to consist of only Republicans or only Democrats.

Sampling

Design

A sample that produces data that is not representative because of the systematic under- or over-estimation of the values of the variable of interest is called *biased*. **Bias may result from either a poor sampling plan or from a poor design for evaluating the variable of interest.**





Probability Sampling Plan

1. **Simple Random Sampling** – the simplest probability sampling plan. It is equivalent to “selecting names out of a hat.” Each individual has the same chance of being selected.
2. **Cluster Sampling** – This sampling technique is used when our population is naturally divided into groups (which we call clusters). For example, all the students in a university are divided into majors. In cluster sampling, we take a random sample of clusters, and **use all the individuals within the selected clusters as our sample**.
 - ❑ Example - in order to get a sample of **high-school seniors** from a certain city, you **choose 3 high schools at random from among all the high schools in that city**, and use **all the high school seniors in the three selected high schools as your sample**.
3. **Stratified Sampling** – when population is naturally divided into sub-populations, which we call stratum. For example, all the students in a certain college are divided by gender or by year in college. In stratified sampling, **we choose a simple random sample from each stratum**, and our sample consists of all these simple random samples put together.
 - ❑ Example - in order to get a random sample of high-school seniors from a certain city, we choose a **random sample of 25 seniors from each of the high schools in that city**. Our sample consists of all these samples put together.

Cluster vs. Stratified sampling - we want to conduct a study on the sleeping habits of UG students at HU, and need to obtain a sample.

- ❑ Cluster sampling - we would randomly choose, say, 2 majors (groups) out of 6, and use all the students in these 2 majors as our sample.
- ❑ Stratified sampling - we would obtain a random sample of, say, 50 students from each of the 6 majors (groups), and use the 300 chosen students as the sample.

Clearly in this example, stratified sampling is much better, since the major of the student might have an effect on the student's sleeping habits, and so we would like to make sure that we have representatives from all the different majors.



Probability Sampling Plan

1. **Cluster Sampling** – This sampling technique is used when our population is naturally divided into groups (which we call clusters). For example, all the students in a university are divided into majors. In cluster sampling, we take a random sample of clusters, and **use all the individuals within the selected clusters as our sample**.
 - ❑ Example - in order to get a sample of **high-school seniors** from a certain city, you **choose 3 high schools at random from among all the high schools in that city**, and use **all the high school seniors in the three selected high schools as your sample**.

Multistage Sampling ?

Multistage sampling can have more than 2 stages. For example, to obtain a random sample of physicians in the United States, you choose 10 states at random (stage 1, cluster). From each state you choose at random 8 hospitals (stage 2, cluster). Finally, from each hospital, you choose 5 physicians from each sub-specialty (stage 3, stratified).



Sample Size

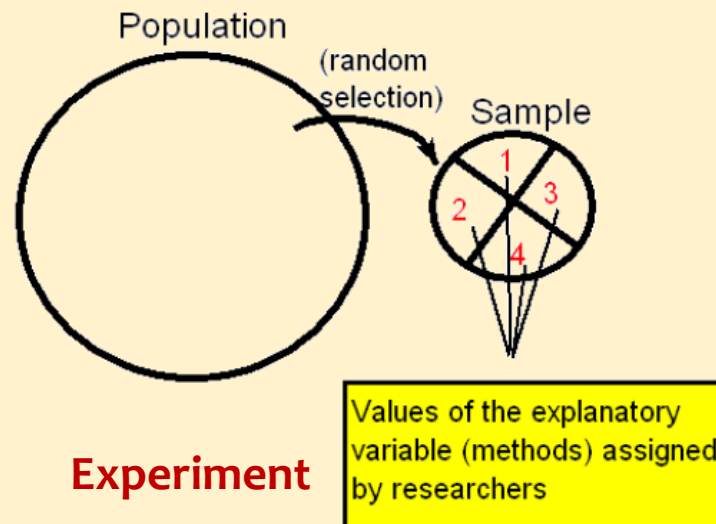
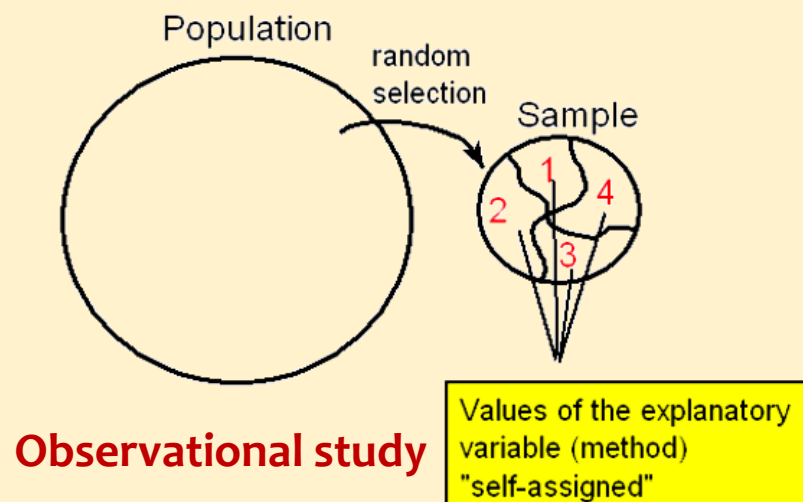
1. To ensure the sample is representative of the population, by using some form of probability sampling plan.
2. To get a more precise idea of what values are taken by the variable of interest for the entire population, ***a larger sample does a better job than a smaller one***. We will discuss the issue of sample size in more detail in the Inference unit and see *how changes in the sample size affect the conclusions we can draw about the population*.

Sampling

Design

Design Studies - *Play a crucial role in determining our ability to establish evidence of causation*

1. **Observational study** – values of the variable or variables of interest are recorded as they naturally occur. There is no interference by the researchers who conduct the study.
2. **Sample Survey** – a particular type of observational study in which individuals report variables' values themselves, frequently by giving their opinions.
3. **Experiment** – Instead of assessing the values of the variables as they naturally occur, the researchers interfere, and they are the ones who assign the values of the explanatory variable to the individuals. The researchers “take control” of the values of the explanatory variable because they want to see how changes in the value of the explanatory variable affect the response variable. (Note: By nature, any experiment involves at least two variables).



Sampling

Design

Design Studies - Example

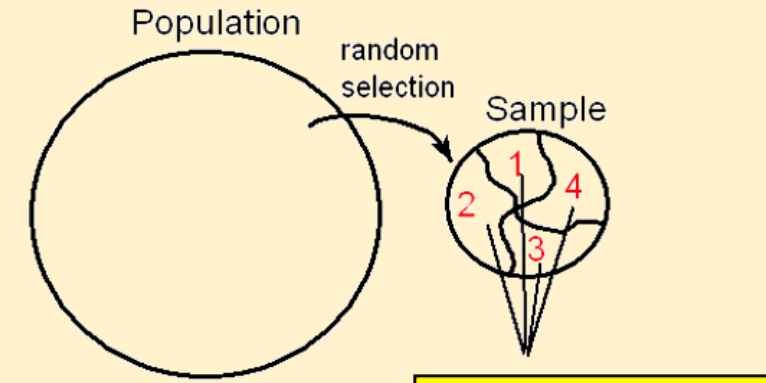
❑ Researchers would like to determine which of the following is the best method for quite smoking:

1. Drugs that alleviate nicotine addiction.
2. Therapy that trains smokers to quit.
3. A combination of drugs and therapy.
4. Neither form of intervention (quitting “cold turkey”).

❑ The explanatory variable is the method (1, 2, 3 or 4), while the response variable is eventual success or failure in quitting.

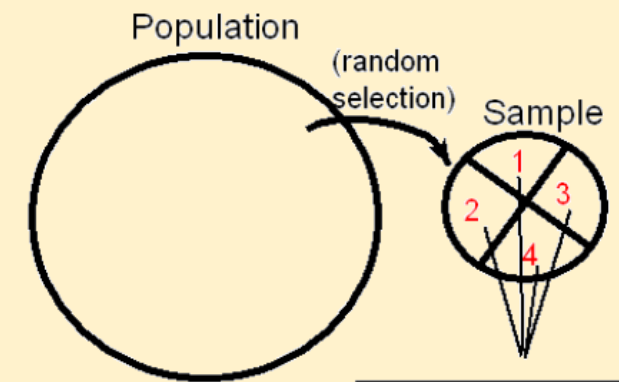
Observation study - In order to obtain a representative sample, we might use a nationwide telephone survey to identify 1,000 smokers who are just beginning to quit smoking. We record which of the four methods the smokers use. One year later, we contact the same 1,000 individuals and determine whether they succeeded.

Experiment - we again collect a representative sample from the population of smokers who are just now trying to quit, using a nationwide telephone survey of 1,000 individuals. This time, however, we divide the sample into 4 groups of 250 and assign each group to use one of the four methods to quit. One year later, we contact the same 1,000 individuals and determine whose attempts succeeded while using our designated method.



Observational study

Values of the explanatory variable (method) "self-assigned"



Experiment

Values of the explanatory variable (methods) assigned by researchers

Sampling

Design

Design Studies - Example

Last time!

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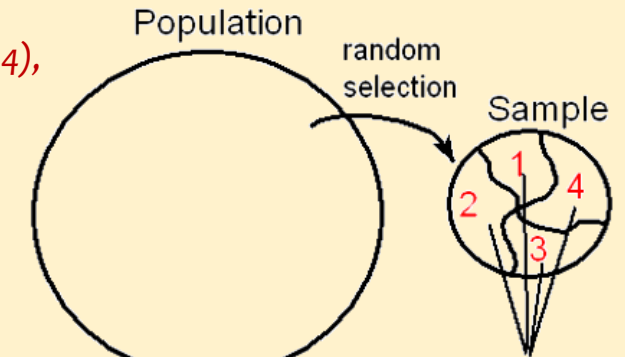
❑ Explanatory variables – the 4 methods (1, 2, 3 or 4),

❑ Response variable - success or failure in quitting

	Quit	Didn't Quit	Total	% Who Quit
Cold Turkey	12	238	250	5 %
Drugs only	60	190	250	24 %
Therapy only	59	191	250	24 %
Drugs & Therapy	83	167	250	33 %

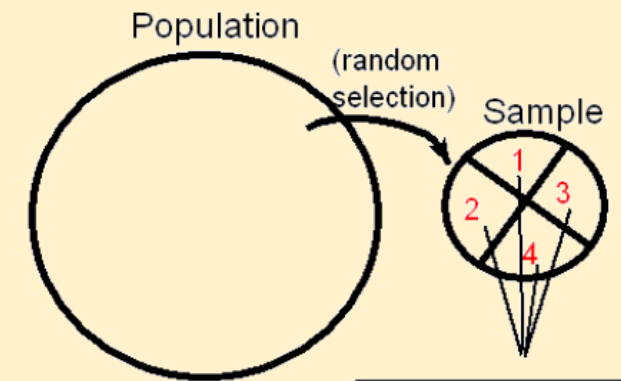
Can we conclude that using the combination drugs and therapy method caused the smokers to quit most successfully?

Which type of design was implemented will play an important role in the answer to this question.



Observational study

Values of the explanatory variable (method) "self-assigned"



Experiment

Values of the explanatory variable (methods) assigned by researchers

Sampling

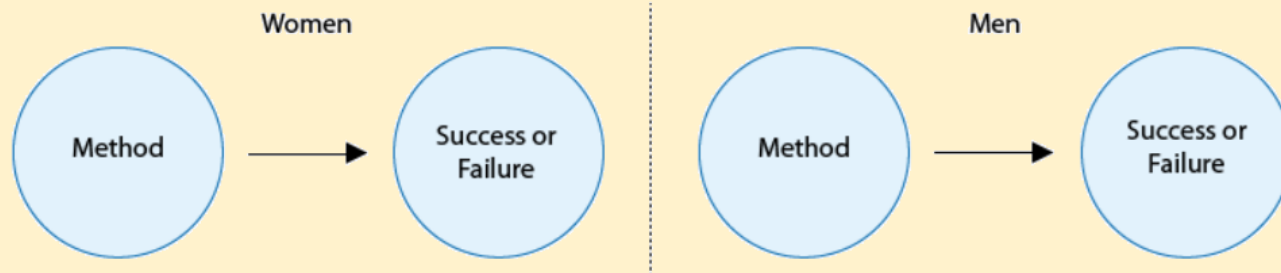
Design

Design Studies - Example

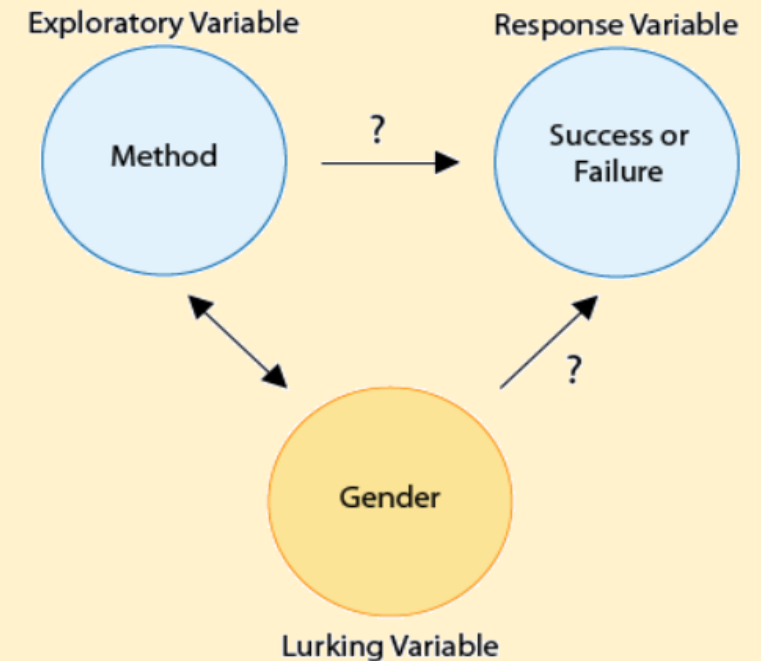
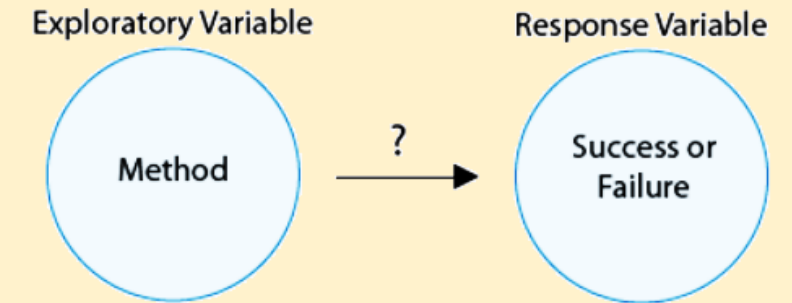
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Causation & Observational study



Sampling

Design

Design Studies - Example

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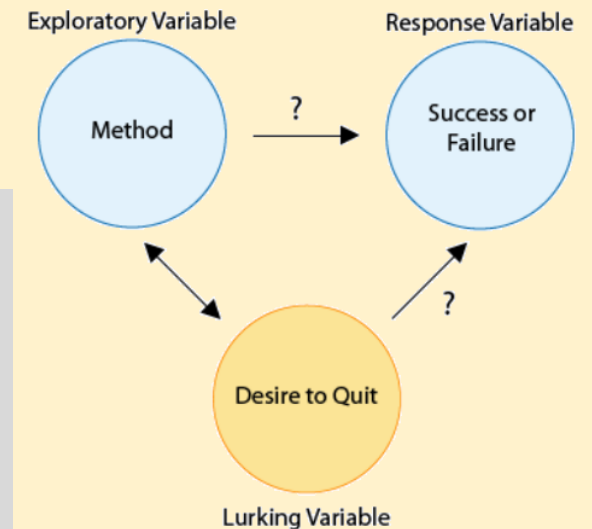
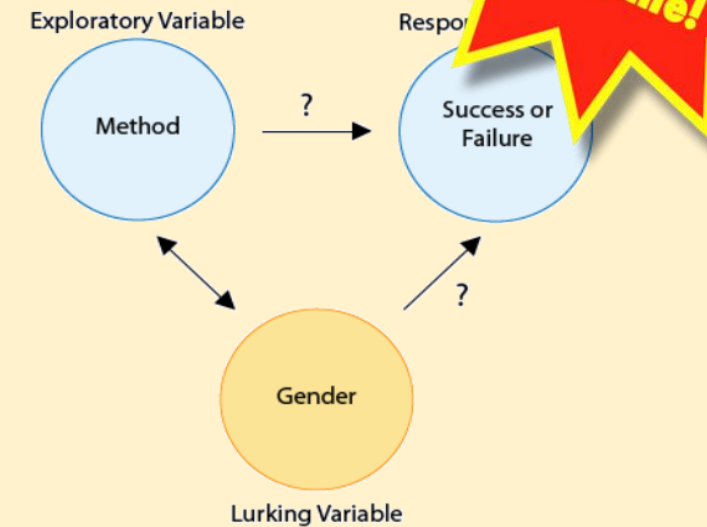
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Causation & Observational study

Establishing ‘causation’ is usually difficult – or perhaps even impossible – in an observational study, especially due to the problem of lurking variables.

The key to establishing causation is to rule out the possibility of any lurking variable, or in other words, **to ensure that individuals differ only with respect to the values of the explanatory variable.**

In general, this is a goal which we have a much better chance of accomplishing by carrying out a well-designed experiment.



Sampling

Design

Design Studies - Example

❑ Researchers would like to determine which of the following is the best method:

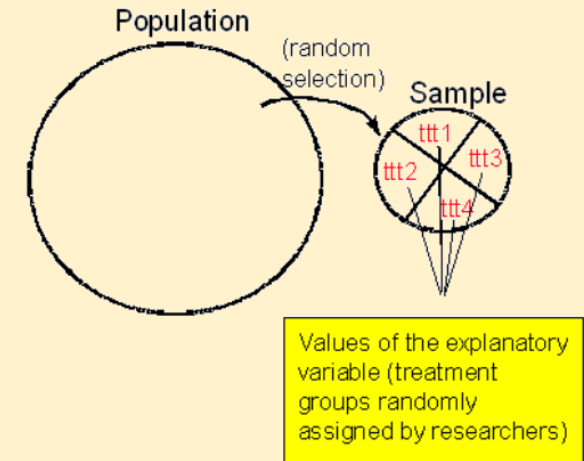
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Can we conclude that using the combination drugs and therapy method caused the smokers to quit most successfully?

Key to ensuring that individuals differ only with respect to explanatory values – which is also the key to establishing causation – lies in the way this assignment is carried out

Causation & Experiment



1. Randomized Controlled Experiments
2. Inclusion of a Control Group
3. Blind and Double-Blind Experiments



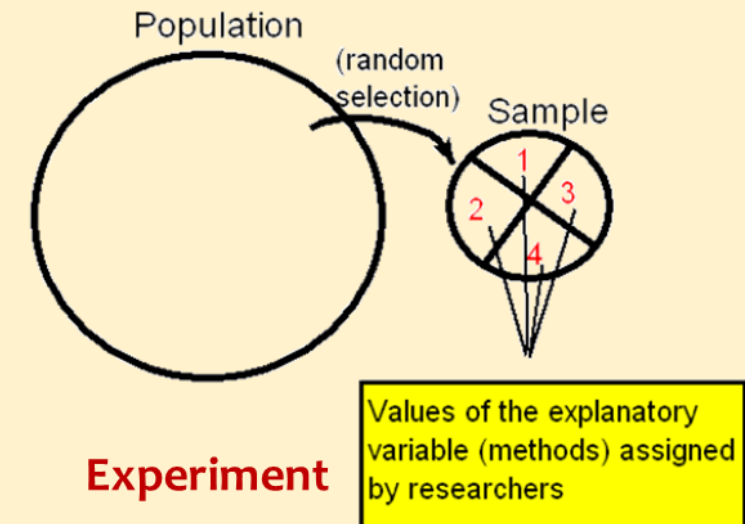
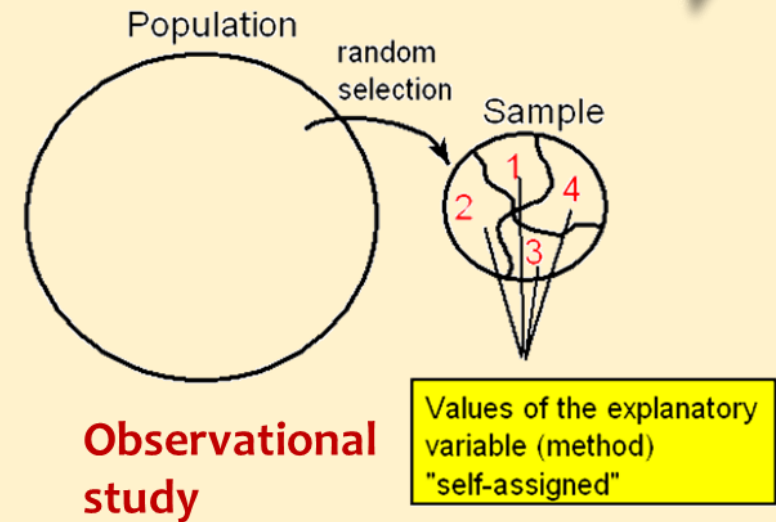
Summary – Observational Study vs. Experiment

1. Observational studies:

- ☐ The explanatory variable's values are allowed to occur naturally.
- ☐ Because of the possibility of lurking variables, it is difficult to establish causation.
- ☐ If possible, control for suspected lurking variables by studying groups of similar individuals separately.
- ☐ Some lurking variables are difficult to control for; others may not be identified.

2. Experiments:

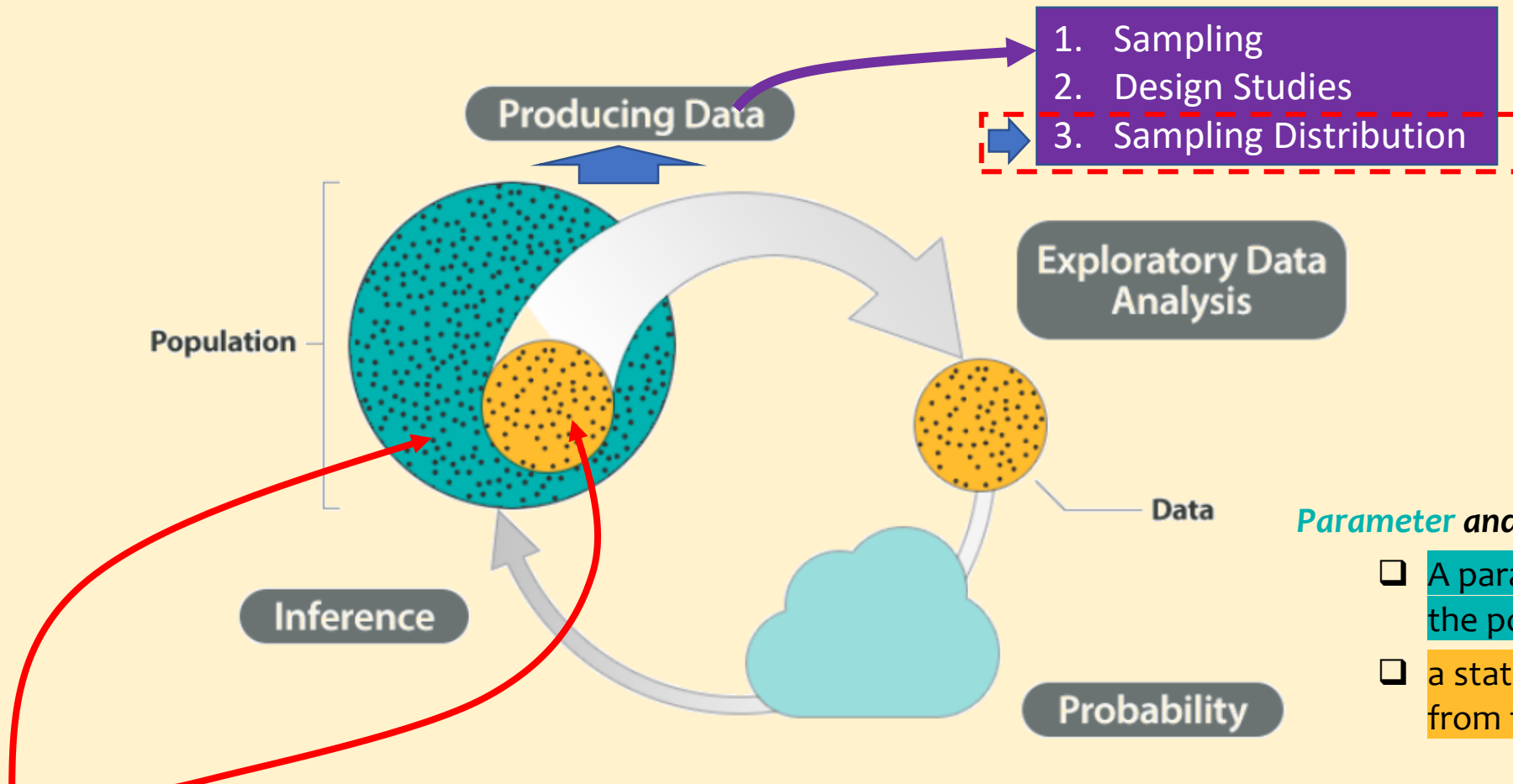
- ☐ The explanatory variable's values are controlled by researchers (treatment is imposed).
- ☐ Randomized assignment to treatments automatically controls for all lurking variables.
- ☐ Making subjects blind avoids the placebo effect.
- ☐ Making researchers blind avoids conscious or subconscious influences on their subjective assessment of responses.
- ☐ A randomized controlled double-blind experiment is generally optimal for establishing causation.





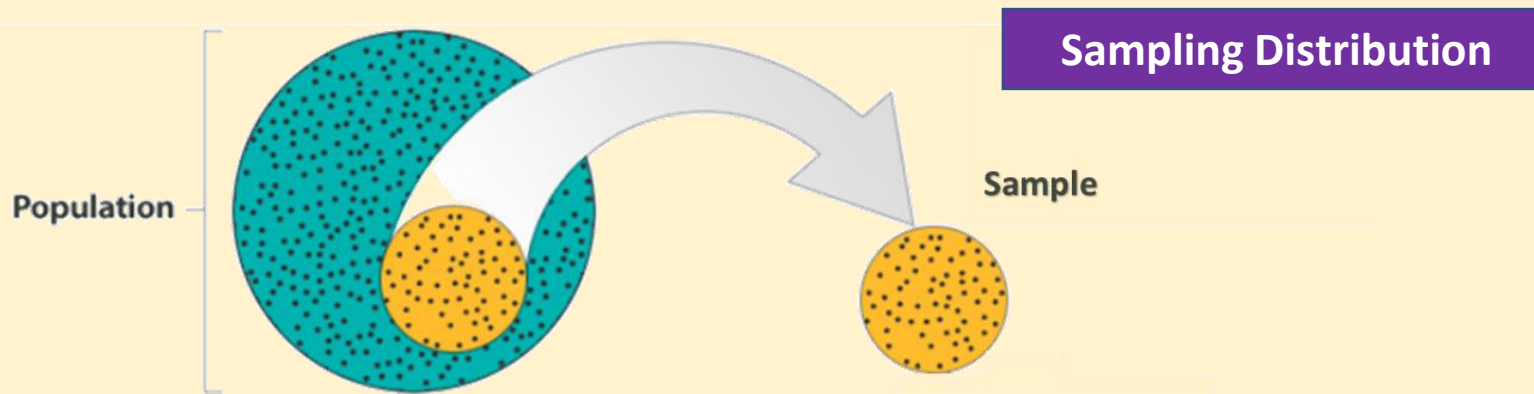
Last time!

Sample surveys are occasionally used to examine relationships, but often they assess values of many separate variables, such as respondents' opinions on various matters. Survey questions should be designed carefully, in order to ensure unbiased assessment of the variables' values.



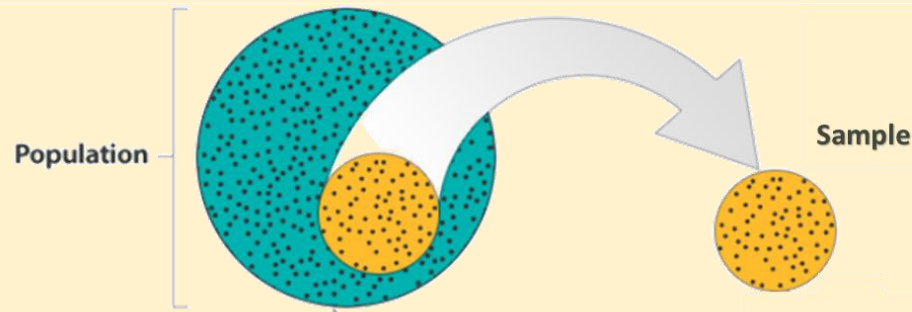
The Law of Large Numbers states that as the number of trials increases, the relative frequency becomes the actual probability. So, using this law, as the number of trials increases, the empirical probability gets closer and closer to the theoretical probability.

Law of Large Numbers: The actual (or true) probability of an event (A) is estimated by the relative frequency with which the event occurs in a long series of trials.



Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

- ❑ Parameters are usually unknown, because it is impractical or impossible to know exactly what values a variable takes for every member of the population.
- ❑ Statistics are computed from the sample, and vary from sample to sample due to *sampling variability*.
- ❑ In today's lecture, **statistical inference**, we will learn how to use a statistic to draw conclusions about an unknown parameter, either by estimating it or by deciding whether it is reasonable to conclude that the parameter equals a proposed value.
- ❑ Here, we'll learn about the **behavior of the statistics assuming that we know the parameters**.
 - ❑ Example - if we know that the population proportion of blood type A in the population is 0.42, and we take a random sample of size 500, **what do we expect the sample proportion $\hat{p}$ to be?**



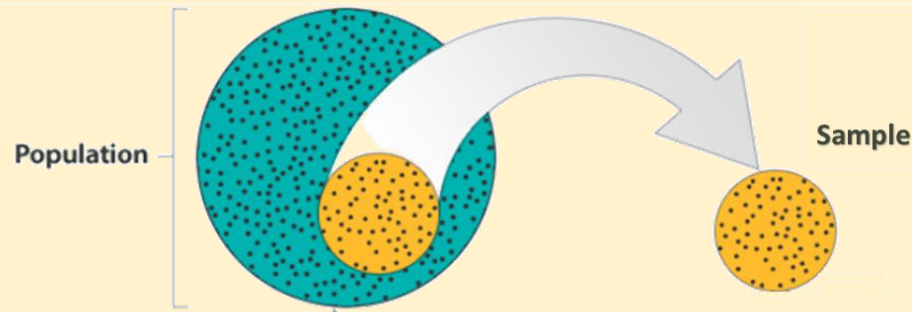
Sampling Distribution

Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
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Behavior of sample proportion, $\hat{p}$

- ❑ Due to sampling variability - sample proportion, $\hat{p}$ is a random variable
- ❑ To understand $\hat{p}$, we usually analyze – 1) the center, 2) the spread, and 3) the shape
 - ❑ **Center** - the mean of the distribution of $\hat{p} = p$
 - ❑ **Spread** - sample size plays a role in the spread of the distribution of sample proportion: **there should be less spread for larger samples, more spread for smaller samples**
 - ❑ **Shape** - shape of the distribution of sample proportion should bulge in the middle and taper at the ends: **it should be somewhat normal.**

❑ Sampling Distribution - distribution of the values of the sample proportions $\hat{p}$ in repeated samples is called the sampling distribution of $\hat{p}$



Sampling Distribution

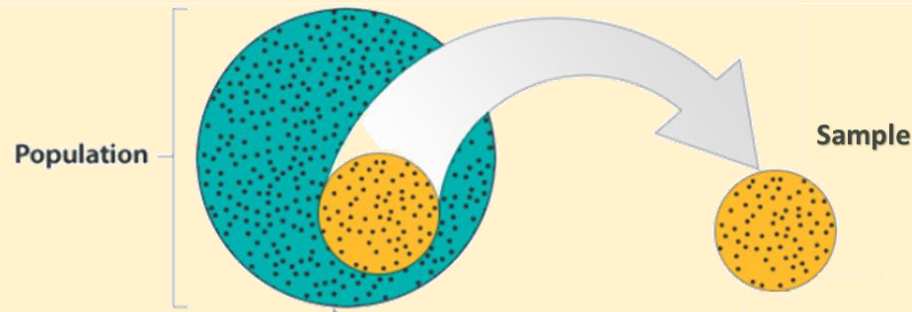
Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
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Behavior of sample proportion, $\hat{p}$

- Sampling Distribution - distribution of the values of the sample proportions $\hat{p}$ in repeated samples is called the sampling distribution of $\hat{p}$

$\hat{p}$ has a normal distribution with a mean of $\mu_{\hat{p}} = p$ and standard deviation $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ (and as long as np and $n(1-p)$ are at least 10).

example, $n = 25$ (sample size) and $p = 0.6$. Note that $np = 15 \geq 10$ and $n(1-p) = 10 \geq 10$. Therefore we can conclude that $\hat{p}$ is approximately a normal distribution with mean $p = 0.6$ and standard deviation $\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.6(1-0.6)}{25}} = 0.097$



Sampling Distribution

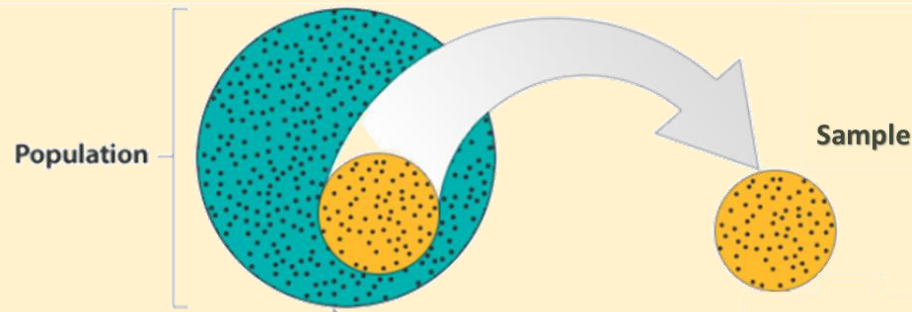
Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s



Behavior of sample mean, $\bar{x}$

- Sampling Distribution - distribution of the values of the sample mean ($\bar{x}$) in repeated samples is called the sampling distribution of $\bar{x}$

The Sampling Distribution of the Sample Mean If repeated random samples of a given size n are taken from a population of values for a quantitative variable, where the population mean is μ and the population standard deviation is σ , then the mean of all sample means ($\bar{x}$) is population mean μ . As for the spread of all sample means, theory dictates the behavior much more precisely than saying that there is less spread for larger samples. In fact, the standard deviation of all sample means ($\bar{x}$) is exactly $\frac{\sigma}{\sqrt{n}}$. Since the square root of sample size n appears in the denominator, the standard deviation does decrease as sample size increases.

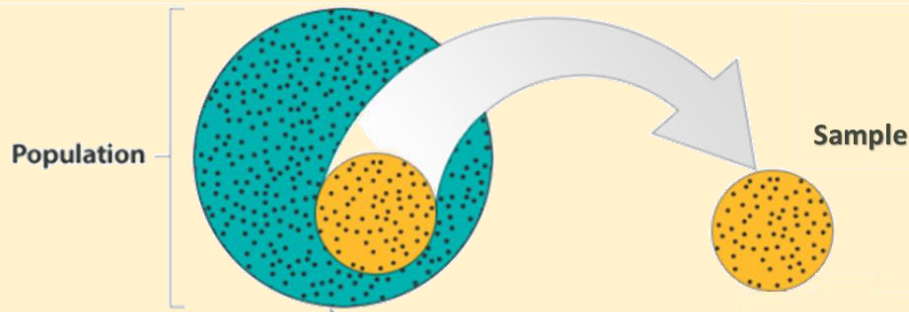


Sampling Distribution

Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

Sampling distributions for sample means and sample proportions

			Sampling Distribution		
Variable	Parameter	Statistic	Center	Spread	Shape
Categorical (example: left-handed or not)	p = population proportion	$\hat{p}$ = sample proportion	p	$\sqrt{\frac{p(1-p)}{n}}$	Normal IF $np \geq 10$ and $n(1-p) \geq 10$
Quantitative (example: age)	μ = population mean, σ = population standard deviation	$\bar{x}$ = sample mean	μ	$\frac{\sigma}{\sqrt{n}}$	When will the distribution of sample means be approximately normal ?



Summary - Sampling Distribution

Numerical Measure	(Population) Parameter	(Sample) Statistic
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

Sampling distributions for sample means and sample proportions

- ❑ Statistics vary from sample to sample due to sampling variability, and therefore can be regarded as *random variables* whose distribution we call sampling distribution
 - ❑ Sample proportion, $\hat{p}$, and the sample mean, $\bar{x}$ relative to their respective population parameters (p & μ)
 - ❑ under certain conditions the sampling distribution is approximately normal. This result is known as the Central Limit Theorem - the foundation for statistical inference

WS 26

Q20: ... including children, are **injured or killed** ...

Q21: Which of the following is a **statistic**?

Q22: Which of the following is a **parameter**?

Q23: **x , μ , s** , and **σ** are:

Q24: Compared to small samples, large samples have { less, more, about the same } **variability**

Q25: Which of the following is a plausible standard deviation for samples of **100**?

Q26: For which of the following sample sizes is a **normal** model a good fit for the sampling distribution of sample proportions?

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Sample Surveys

 Question No 20 – Consider this survey question: “Every year, thousands of people, including children, are **injured or killed** because of the widespread availability of firearms in the United States. Do you agree that our country needs stricter gun control laws?” The response that surveyors hope to elicit is:

- A. Yes
- B. No
- C. Unknown

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – Sample Surveys

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- A. Yes
- B. No
- C. Unknown

Answer A.

The question is worded in a way that highlights negative facts about the availability of guns.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Parameters vs. Statistics

Background - The SAT-Verbal scores of a sample of 300 students at a particular university had a mean of 592 and standard deviation of 73. According to the university's reports, the SAT-Verbal scores of all its students had a mean of 580 and a standard deviation of 110.

 **Question No 21 – Which of the following is a **statistic**?**

- A. 300.
- B. 592.
- C. 580.
- D. 110.

Reading Exercise – No 26

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 **Question No 21 – Which of the following is a statistic?**

- A. 300.
- B. 592.**
- C. 580.
- D. 110.

Answer B.

592 is the sample mean, which is a statistic.

Reading Exercise – No 26

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 **Question No 22 – Which of the following is a **parameter**?**

- A. 300.
- B. 592.
- C. 110.
- D. 73.

Reading Exercise – No 26

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 **Question No 22 – Which of the following is a **parameter**?**

- A. 300.
- B. 592.
- C. 110.**
- D. 73.

Answer C.

110 is the population standard deviation, which is a parameter.

Reading Exercise – No 26

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 **Question No 23** – $\bar{x}$, μ , s , and σ are?

- A. 73, 110, 592, 580.
- B. 592, 580, 73, 110.
- C. 580, 592, 110, 73.
- D. 300, 592, 73, 110.

Reading Exercise – No 26

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 **Question No 23** – $\bar{x}$, μ , s , and σ are?

- A. 73, 110, 592, 580.
- B. 592, 580, 73, 110.**
- C. 580, 592, 110, 73.
- D. 300, 592, 73, 110.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Behavior of Sample Proportion

 **Question No 24** – Compared to small samples, large samples have {less, more, about the same} **variability**.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Behavior of Sample Proportion

 **Question No 24** – Compared to small samples, large samples have {less, more, about the same} **variability**.

Less. In the simulation, the sample proportions for $n = 100$ were more tightly grouped about the population proportion.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Behavior of Sample Proportion

 **Question No 25** – We collect random samples of 25 students at a time and calculate the proportion of females in each sample. The standard deviation of $\hat{p}$ is approximately 0.10. Which of the following is a plausible standard deviation for samples of 100?

- A. 0.40.
- B. 0.10.
- C. 0.05.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Behavior of Sample Proportion

 **Question No 25** – We collect random samples of 25 students at a time and calculate the proportion of females in each sample. The standard deviation of $\hat{p}$ is approximately 0.10. Which of the following is a plausible standard deviation for samples of 100?

- A. 0.40.
- B. 0.10.
- C. 0.05.

Answer C.

If the sample size is increased, the standard deviation will decrease because larger samples have less variability.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Behavior of Sample Proportion

 **Question No 26** – According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, 62% of graduates from public universities had student loans. We randomly sample college graduates from public universities and determine the proportion in the sample with student loans. For which of the following sample sizes is a **normal model** a good fit for the sampling distribution of sample proportions?

- A. 10.
- B. 20.
- C. 30.
- D. Both 20 and 30.
- E. None of these.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Behavior of Sample Proportion 2

 **Question No 26** – According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, 62% of graduates from public universities had student loans. We randomly sample college graduates from public universities and determine the proportion in the sample with student loans. For which of the following sample sizes is a **normal model** a good fit for the sampling distribution of sample proportions?

- A. 10.
- B. 20.
- C. 30.**
- D. Both 20 and 30.
- E. None of these.

Answer C.

Are $np \geq 10$ and $n(1 - p) \geq 10$?

Both conditions are met. $np = (30)(0.62) = 18.6$ and $n(1 - p) = (30)(0.38) = 11.4$. Both are greater than 10. So a normal model is a good fit for the sampling distribution of sample proportions.

Q27: ...random samples of 30 students?

Q28: ... have more than 40% blue M&M's?

Q29: ... random sample of 50 M&M's ...

Q30: -

Q31: What interval is almost certain ...

Q32: ... sample of 225 people

Q33: -

Q34: randomly sample 36 Pell grant recipients

Q35: What is (the) mean of the sample means

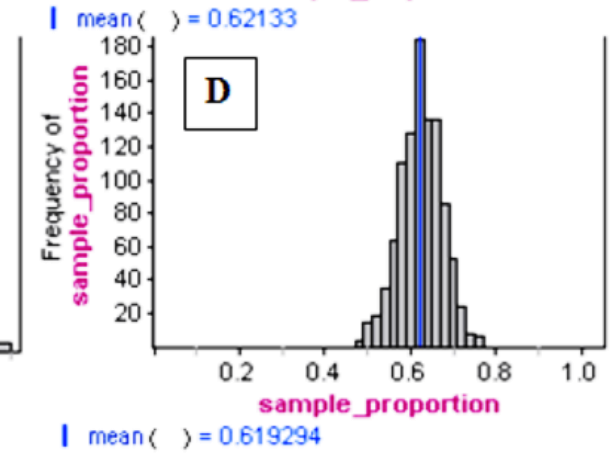
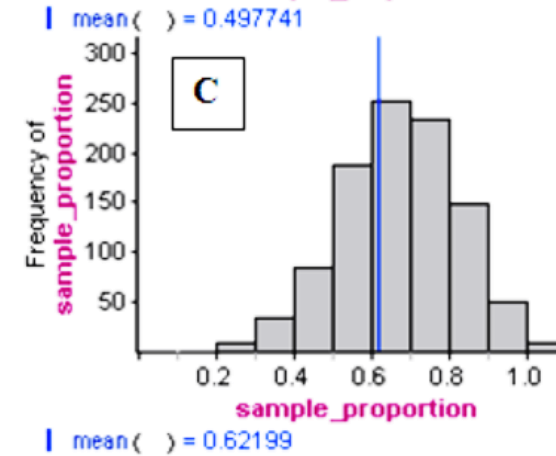
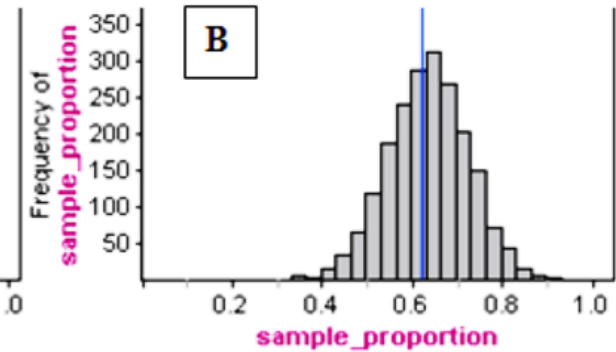
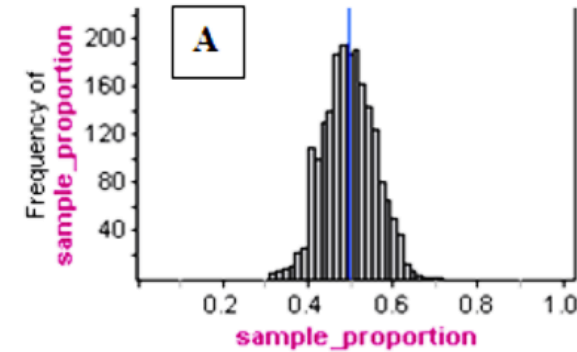
Q36: What is the standard deviation of the sample means?

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Behavior of Sample Proportion 2

Question No 27 – Which distribution is a plausible representation of the sampling distribution for random samples of **30 students**?



Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Behavior of Sample Proportion 2

Question No 27 – Which distribution is a plausible representation of the sampling distribution for random samples of **30 students**?

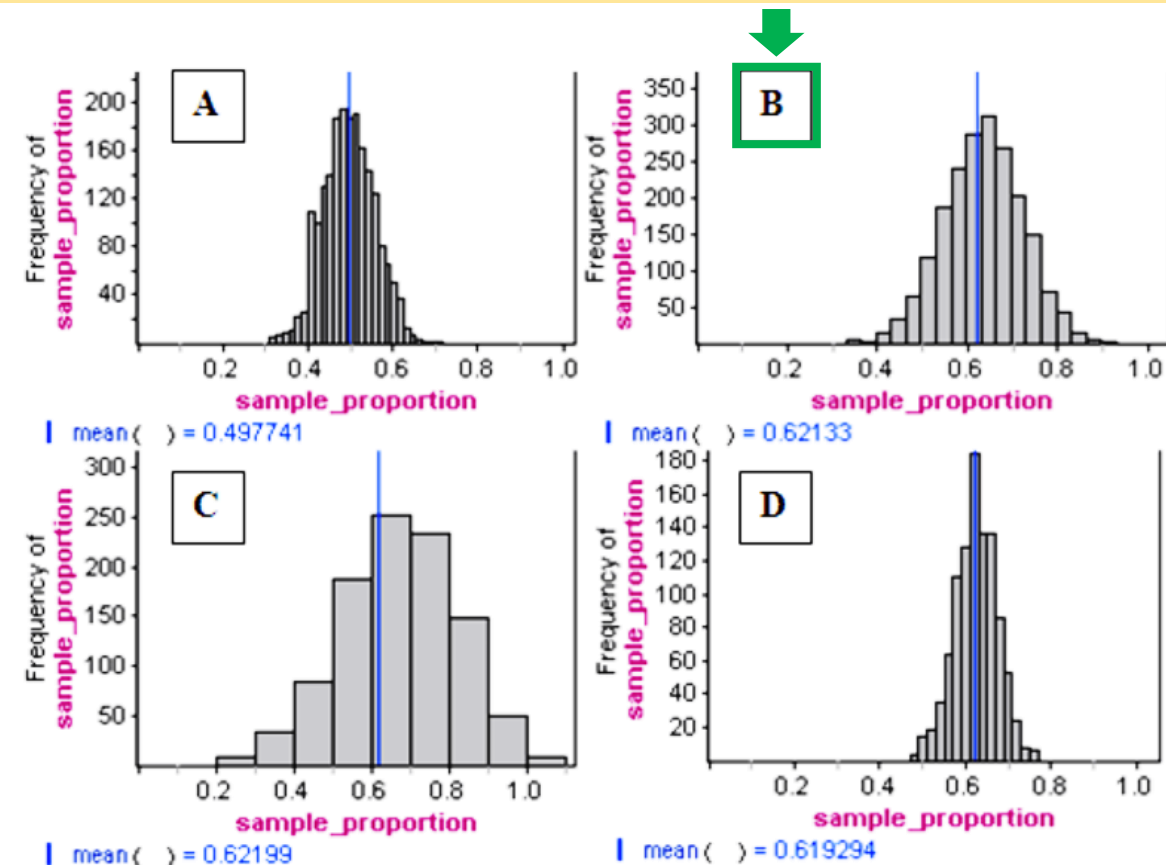
Answer B.

The distribution of sample proportions has a mean of p and a standard deviation of $\sqrt{p(1-p)/n}$.

p is the population proportion. In this scenario, p is the proportion of graduates from public universities who have student loans. n is the sample size.

Plug in $p = 0.62$ and $n = 30$. Look for a graph with a mean of approximately 0.62 and standard deviation of approximately 0.09. We know from the Standard Deviation Rule that roughly 68% of the sample proportions will fall within 1 standard deviation of the mean.

The mean of the sampling distribution should be $p = 0.62$ with standard deviation $\sqrt{p(1-p)/n} = \sqrt{0.62(1-0.62)/30} = 0.09$. Typical values should fall within one standard deviation of the mean, from about 0.53 to 0.71. This distribution fits this description, as shown in the graph.



Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Behavior of Sample Proportion 3

 **Question No 28** – According to the official M&M website, 24% of the plain milk chocolate M&M's produced by Mars Corporation are blue. Annie buys a large family-size bag of M&M's. Sarah buys a small fun-size bag.

Which bag is more likely to have more than 40% blue M&M's?:

- A. Annie, because there are more M&M's in her bag, so she will have more blue ones.
- B. Annie, because there is more variability in the proportion of blues among larger samples.
- C. Sarah, because there is more variability in the proportion of blues among smaller samples.
- D. Both have the same chance because they are both random samples.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Behavior of Sample Proportion 3

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- B. Annie, because there is more variability in the proportion of blues among larger samples.
- C. Sarah, because there is more variability in the proportion of blues among smaller samples.**
- D. Both have the same chance because they are both random samples.

Answer C.

Think about the spread of the sampling distribution around $p = 0.24$.

For which size of bag will sample proportions greater than 0.40 be more common?

If $\hat{p} > 0.40$ occur frequently, the sampling distribution has a large spread around $p = 0.24$. In the sampling distribution, will we see more variability when using a large sample size or a small sample size?

There is more variability in small samples, so it is more likely to get sample results further from $p = 0.24$ with a small bag.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Behavior of Sample Proportion 3

 **Question No 29** – Imagine that you have a very large barrel that contains tens of thousands of M&M's. According to the official M&M website, 20% of the M&M's produced by the Mars Corporation are orange.

5 students each take a random sample of **50** M&M's and record the percentage of orange in each sample. **Which sequence is the most plausible for the percentage of orange candies obtained in these 5 samples?**

- A. 20%, 20%, 20%, 20%, 20%.
- B. 15%, 25%, 22%, 20%, 28% .
- C. 5%, 80%, 8%, 65%, 70%.
- D. Any of the above.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Behavior of Sample Proportion 3

 **Question No 29** – Imagine that you have a very large barrel that contains tens of thousands of M&M's. According to the official M&M website, 20% of the M&M's produced by the Mars Corporation are orange.

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- A. 20%, 20%, 20%, 20%, 20%.
- B. 15%, 25%, 22%, 20%, 28% .
- C. 5%, 80%, 8%, 65%, 70%.
- D. Any of the above.

Answer B

Think about the center and spread in the sampling distribution. Which sample results are the most likely to occur? (Of course, all of these are possible. Which is the most plausible?)

The mean of the sampling distribution is 0.20. The standard deviation is $\sqrt{0.2(1-0.2)/50} = 0.06$. Which sequence can we rule out because most of the p-hats are more than 1 standard deviation from $p = 0.20$? Of the sequences left, which is the most plausible?

We expect p-hats within 1 standard deviation of $p = 0.20$ to be most common. The standard deviation is about 0.06/ 4 of the 5 p-hats in this sequence are within 0.06 of $p = 0.20$.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 **Question No 30 – What is the sampling distribution of the sample proportion ($\hat{p}$)?** In other words, what can we say about the behavior of the different possible values of the sample proportion that we can get when we take such a sample?

(Note: normal approximation is valid because $0.1(225) = 22.5$ and $0.9(225) = 202.5$ are both more than 10.)

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 **Question No 30 – What is the sampling distribution of the sample proportion ($\hat{p}$)?** In other words, what can we say about the behavior of the different possible values of the sample proportion that we can get when we take such a sample?

(Note: normal approximation is valid because $0.1(225) = 22.5$ and $0.9(225) = 202.5$ are both more than 10.)

Answer

The possible values of the sample proportion follow approximately a normal distribution with mean $p = 0.1$, and standard deviation =

$$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.1(1-0.1)}{225}} = 0.02.$$

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 Question No 31 – What **interval** is almost certain (probability 0.997) to contain the sample proportion of left-handed people?

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 Question No 31 – What **interval** is almost certain (probability 0.997) to contain the sample proportion of left-handed people?

Answer

Since the Standard Deviation Rule applies, the probability is approximately 0.997 that the sample proportion falls within 3 standard deviations of its mean, that is, between $0.1 - 3(0.02)$ and $0.1 + 3(0.02)$. There is roughly a 99.7% chance, therefore, that the sample proportion falls in the interval $(0.04, 0.16)$.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.



Question No 32 – In a sample of 225 people, would it be unusual to find that 40 people in the sample are left-handed?

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 **Question No 32 – In a sample of 225 people, would it be unusual to find that 40 people in the sample are left-handed?**

Answer - About 18% ($40/225$) of this sample is left-handed. In the previous problem, we determined that there is roughly a 99.7% chance that a sample proportion will fall between 0.04 and 0.16. So a sample proportion of 0.18 is very unlikely. Note: According to the Standard Deviation Rule, sample proportions greater than 0.16 will occur 0.15% of the time. $(100\% - 99.7\%) / 2 = 0.15\%$.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 **Question No 33 – Find the approximate probability of at least 27 in 225 (proportion 0.12) being left-handed. In other words, what is $P(\hat{p} \geq 0.12)$?**

Guidance: Note that 0.12 is exactly 1 standard deviation (0.02) above the mean (0.1). Now use the Standard Deviation Rule.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Behavior of Sample Proportion 3

The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

 **Question No 33 – Find the approximate probability of at least 27 in 225 (proportion 0.12) being left-handed. In other words, what is $P(\hat{p} \geq 0.12)$?**

Guidance: Note that 0.12 is exactly 1 standard deviation (0.02) above the mean (0.1). Now use the Standard Deviation Rule.

Answer - Note that 0.12 is exactly 1 standard deviation above the mean. The Standard Deviation Rule tells us that there is a 68% chance that the sample proportion falls within 1 standard deviation of its mean, that is, between 0.08 and 0.12. There is therefore a probability of $(1 - 0.68) / 2 = 0.16$ that the sample proportion falls above 0.12.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grant awards was \$500.

 **Question No 34 – If we randomly sample 36 Pell grant recipients and record the mean Pell grant award for the sample, then repeat the sampling process many, many times, what is the mean and standard deviation of the sample means?**

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grant awards was \$500.

 **Question No 34 – If we randomly sample 36 Pell grant recipients and record the mean Pell grant award for the sample, then repeat the sampling process many, many times, what is the mean and standard deviation of the sample means?**

Answer - The mean of the distribution of sample means will be μ . The standard deviation will be $\sigma/\sqrt{n}$, where σ is the standard deviation of the population.

$\mu = 2,600$, the average Pell grant for 2008.

$\sigma = 500$, the standard deviation in Pell grants for 2008.

$n = 36$, the sample size.

The distribution of sample means will have a mean equal to $\mu = 2,600$ and a standard deviation of $\sigma/\sqrt{n} = 500/\sqrt{36} = 83.3$.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The population mean for Verbal IQ scores is 100, with a standard deviation of 15. Suppose a researcher takes 50 random samples with 30 people in each sample.

 Question No 35 – What is **mean** of the sample means?

- A. 100.
- B. 15.
- C. 50.
- D. 30.
- E. Unable to determine from the information provided.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The population mean for Verbal IQ scores is 100, with a standard deviation of 15. Suppose a researcher takes 50 random samples with 30 people in each sample.

 Question No 35 – What is **mean** of the sample means?

- A. 100.
- B. 15.
- C. 50.
- D. 30.
- E. Unable to determine from the information provided.

Answer - A.

The mean of the sample means is the population mean; therefore, the mean of the sample means or the sampling distribution of the mean is 100.

Reading Exercise – No 26

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The population mean for Verbal IQ scores is 100, with a standard deviation of 15. Suppose a researcher takes 50 random samples with 30 people in each sample.

 **Question No 36 – What is the **standard Deviation** of the sample means?**

- A. 15.
- B. 2.74.
- C. 2.12.
- D. Unable to determine.

Reading Exercise – No 16

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Behavior of Sample Mean

The population mean for Verbal IQ scores is 100, with a standard deviation of 15. Suppose a researcher takes 50 random samples with 30 people in each sample.

 Question No 36 – What is the **standard Deviation** of the sample means?

- A. 15.
- B. 2.74.**
- C. 2.12.
- D. Unable to determine.

Answer - B.

The standard deviation of the sample means is calculated by dividing the population standard deviation by the square root of the sample size; therefore, $\sigma/\sqrt{n} = 15/\sqrt{30} = 15/5.48 = 2.74$.

Reading Exercise – No 16

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Behavior of Sample Mean

Recall our earlier scenario: The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grants awards was \$500.

 **Question No 37 – If we randomly sample 36 Pell grant recipients, would you be surprised if the mean grant amount for the sample was \$2,940? Pick the correct response that gives the best reason.**

- A. No, \$2,940 would not be surprising because this sample result is within 2 standard deviations of the overall mean grant amount of \$2,600.
- B. Yes, \$2,940 would be surprising because this sample result is \$340 greater than the overall mean grant amount of \$2,600.
- C. No, \$2,940 would not be surprising because this sample result is only \$340 greater than the overall mean grant amount of \$2,600, and we expect there to be variability in sample means.
- D. Yes, \$2,940 would be surprising because this sample result is more than 3 standard deviations from the overall mean grant amount of \$2,600.

Reading Exercise – No 16

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Behavior of Sample Mean

Recall our earlier scenario: The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grants awards was \$500.

 **Question No 37 – If we randomly sample 36 Pell grant recipients, would you be surprised if the mean grant amount for the sample was \$2,940? Pick the correct response that gives the best reason.**

- A. No, \$2,940 would not be surprising because this sample result is within 2 standard deviations of the overall mean grant amount of \$2,600.
- B. Yes, \$2,940 would be surprising because this sample result is \$340 greater than the overall mean grant amount of \$2,600.
- C. No, \$2,940 would not be surprising because this sample result is only \$340 greater than the overall mean grant amount of \$2,600, and we expect there to be variability in sample means.
- D. Yes, \$2,940 would be surprising because this sample result is more than 3 standard deviations from the overall mean grant amount of \$2,600.**

Answer - D. Since $n = 36$, the sample means will have an approximately normal distribution with mean \$2,600 and standard deviation of $500/(\sqrt{36})$. How many standard deviations is \$2,940 above \$2,600? \$2,940 is more than 3 standard deviations above \$2,600. (Three standard deviations above 2,600 is $2,600 + 3(500/6) = 2,850$).

Since the sample means are approximately normal, the Standard Deviation Rule predicts that approximately 99.7% of the sample means will fall within 3 standard deviations of $\mu = 2,600$. Samples means outside this range would be very unlikely. For $n=36$, sample means are approximately normal, so we can use the Standard Deviation Rule. Three standard deviations above 2,600 is $2,600 + 3(500/6) = 2,850$. So \$2,940 is more than 3 standard deviations above \$2,600, thus this sample mean would be surprising.

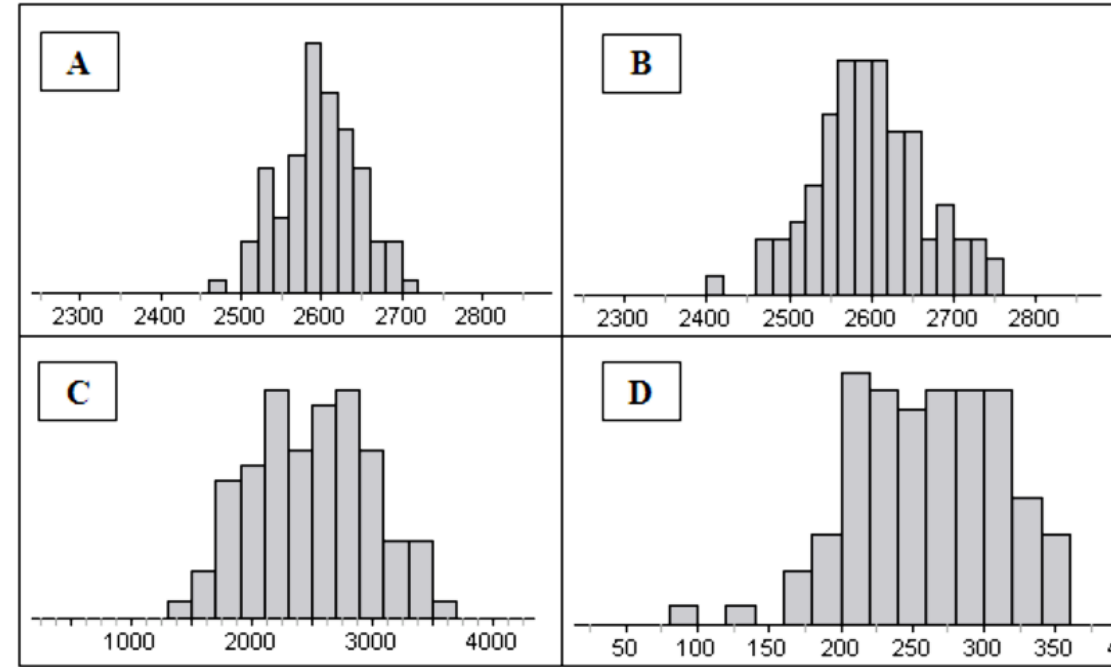
Reading Exercise – No 16

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Behavior of Sample Mean

Recall our earlier scenario: The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grants awards was \$500.

Question No 38 – Which distribution is a plausible representation of 100 samples, with each sample containing 50 randomly selected students?



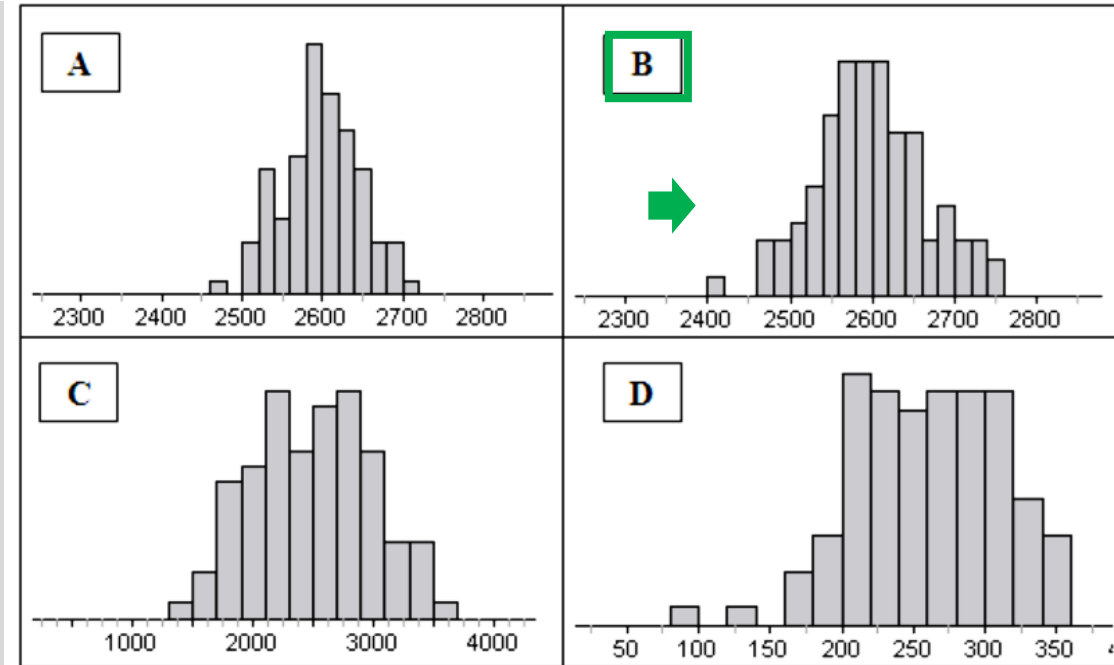
Reading Exercise – No 16

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M - Reflect and discuss – Behavior of Sample Mean

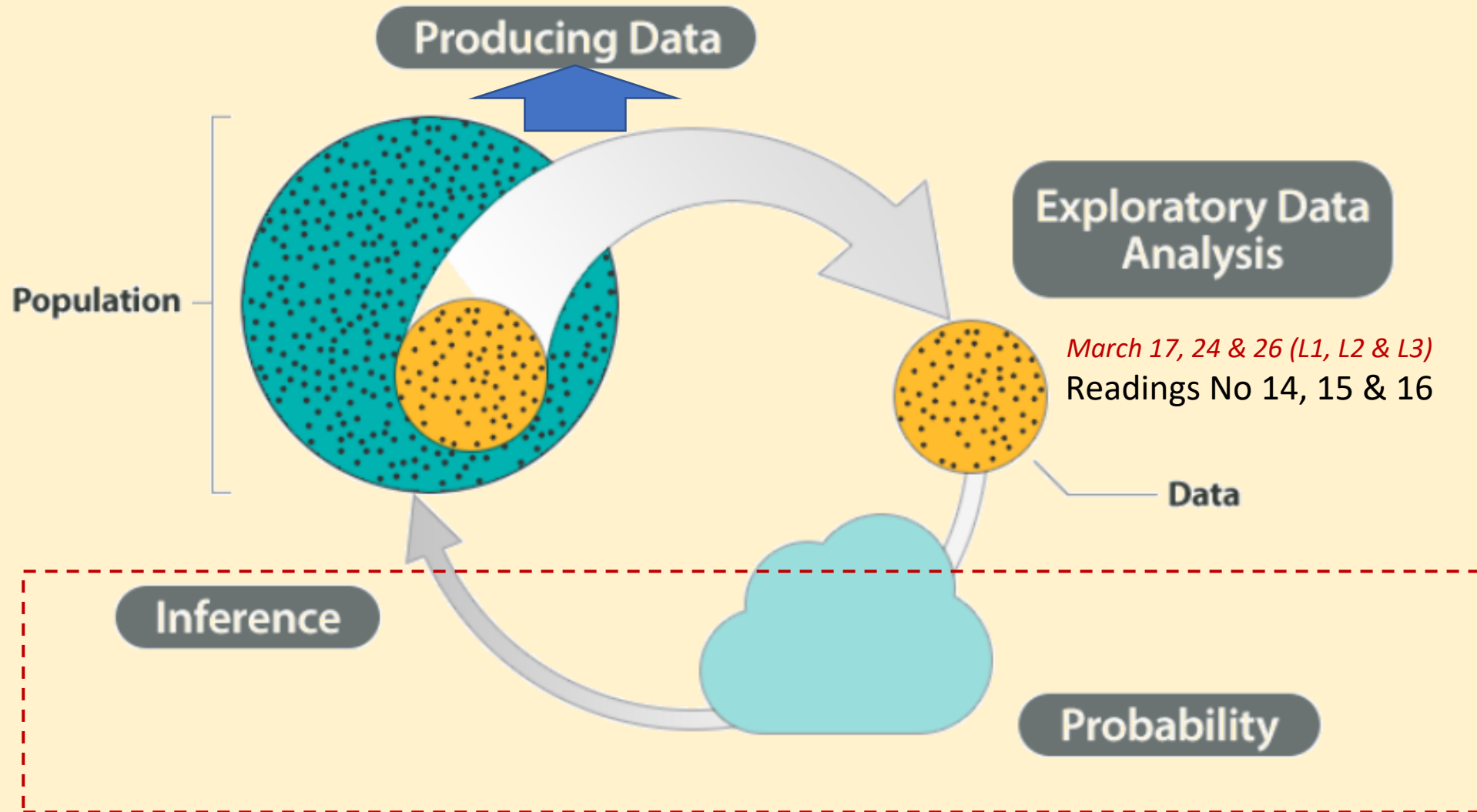
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Question No 38 – Which distribution is a plausible representation of 100 samples, with each sample containing 50 randomly selected students?



Answer - B.

The sampling distribution will have a mean of 2,600 and a standard deviation of 70. Using the Standard Deviation Rule, approximately 68% of the values should be between 2,530 and 2,670 (within 1 standard deviation of the mean.). Graph B looks like it could fit this description.



Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A



Q1. Based on a recent poll of 1,015 U.S. households that have an Internet connection, the researchers claimed with 95% confidence that among all U.S. households that have an Internet connection, between 63% and 67% have a high-speed link.

- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A

 **Q1.** Based on a recent poll of 1,015 U.S. households that have an Internet connection, the researchers claimed with 95% confidence that among all U.S. households that have an Internet connection, between 63% and 67% have a high-speed link.

- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Answer B

The proportion of households with an Internet connection that have a high-speed link is estimated to be (with 95% confidence) somewhere in the interval between 63% and 67%

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A



Q2. A company that provides coaching for the SAT claims in one of its ads: “90% of our students improve their SAT scores after attending our course.” To check the company’s claim, 500 students who took the company’s course were sampled, and it was found that 407 of them (81.4%) actually improved their SAT scores after attending the course. Based on the sample results, we have some serious doubts regarding the accuracy of the company’s claim.

- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A



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- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Answer C

In this case, we have a claim, and we are using the data collected from a sample to assess its accuracy.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A



Q3: In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. Based on the study results, the researchers estimated that the mean time that a male 25 to 35 years of age spends exercising in a week is about 3.5 hours.

- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss A

 **Q3:** In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. Based on the study results, the researchers estimated that the mean time that a male 25 to 35 years of age spends exercising in a week is about 3.5 hours.

- A. point estimation.
- B. interval estimation.
- C. hypothesis testing.

Answer A

Here we are using the data to estimate the mean time (our parameter) using a single number: 3.5 hours.

Reflect and Discuss (B)

- A study on exercise habits used a random sample of 2,540 college students (1,220 females and 1,320 males). The study found the following:
 - 818 of the females in the sample exercise on a regular basis.
 - 924 of the males in the sample exercise on a regular basis.
 - The average time that the 1742 students who exercise on a regular basis ($818 + 924$) spend exercising per week is 4.2 hours.

Q4: female

Q5: females

Q6: a point estimate of 4.2

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B



Q3: What is the point estimate for the proportion of all female college students who exercise on a regular basis? (Numbers are rounded to two decimal places as needed.)

- A. 0.32 B. 0.67 C. 0.69 D. 0.48

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B



Q4: What is the point estimate for the proportion of all female college students who exercise on a regular basis? (Numbers are rounded to two decimal places as needed.)

- A. 0.32 B. 0.67 C. 0.69 D. 0.48

Answer - B

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B

 **Q5.** What proportion of the females in the sample exercise on a regular basis?

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B

 **Q5.** What proportion of the females in the sample exercise on a regular basis?

Of the 1,220 females in the sample, 818 exercise on a regular basis, so the sample proportion is $818 / 1,220 = 0.67$.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B



Q6. Which of the following has a point estimate of 4.2?

- A. The mean time that all college students spend exercising per week.
- B. The mean time that all college students who exercise on a regular basis spend exercising per week.
- C. The percentage of all college students who exercise on a regular basis

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss B



Q6. Which of the following has a point estimate of 4.2?

- A. The mean time that all college students spend exercising per week.
- B. The mean time that all college students who exercise on a regular basis spend exercising per week.
- C. The percentage of all college students who exercise on a regular basis

Answer B

Was the average of 4.2 hours of exercise per week calculated using data from all 2,540 students or only from the 1,742 students who exercise on a regular basis?

The value 4.2 was calculated using data only from the 1,742 students who exercise on a regular basis, and therefore, it is the point estimate for the mean number of exercise hours per week in the population of college students who exercise on a regular basis

Reflect and Discuss (D-E)

- Q7: Which of the following is true regarding x ?
- Q8: Which of the two polls estimates p (the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values) **more accurately**?
- Q9: find a **95% confidence interval ...**

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss C



Q7: A researcher wanted to estimate μ , the mean number of hours that students at a large state university spend exercising per week. The researcher collects data from a sample of 150 students who leave the university gym following a workout. Which of the following is true regarding $\bar{x}$?

- A. It is an unbiased estimate for μ .
- B. It is not an unbiased estimate for μ and probably underestimates μ .
- C. It is not an unbiased estimate for μ and probably overestimates μ .

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss C



Q7: A researcher wanted to estimate μ , the mean number of hours that students at a large state university spend exercising per week. The researcher collects data from a sample of 150 students who leave the university gym following a workout. Which of the following is true regarding $\bar{x}$?

- A. It is an unbiased estimate for μ .
- B. It is not an unbiased estimate for μ and probably underestimates μ .
- C. It is not an unbiased estimate for μ and probably overestimates μ .

Answer C

It is not an unbiased estimator for μ because the sample was not a random sample of 150 students from the entire student body. In addition, students who leave the university gym following a workout are likely students who exercise on a regular basis and therefore tend to exercise more, on average, than students in general.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss E

 **Q8:** Two opinion polls were conducted regarding the cultivation of Western ideals in academia. In particular, both polls were conducted in order to estimate p , the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values.

1. The poll by the Gallow company was based on a random sample of 1,024 Pakistani adults and estimated **the above proportion to be 0.60**.
2. The New Research Center poll was based on a random sample of 2,002 Pakistani adults and estimated the above proportion to be 0.57.

Which of the two polls estimates p (the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values) more accurately?

- A. The Gallow company poll.
- B. The New Research Center poll.
- C. Both polls estimate p with the same accuracy because both are based on a random sample.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss E

 **Q8:** Two opinion polls were conducted regarding the cultivation of Western ideals in academia. In particular, both polls were conducted in order to estimate p , the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values.

1. The poll by the Gallow company was based on a random sample of 1,024 Pakistani adults and estimated **the above proportion to be 0.60**.
2. The New Research Center poll was based on a random sample of 2,002 Pakistani adults and estimated the above proportion to be 0.57.

Which of the two polls estimates p (the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values) more accurately?

- A. The Gallow company poll.
- B. The New Research Center poll.
- C. Both polls estimate p with the same accuracy because both are based on a random sample.

Answer B

The larger the sample the point estimate is based on, the closer it is likely to be to the parameter it estimates.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss E

 **Q9:** A study was done on pregnant women who smoked during their pregnancies. In particular, the researchers wanted to study the effect that smoking has on pregnancy length. A sample of 114 pregnant women who were smokers participated in the study and were followed until the birth of their child. At the end of the study, the collected data were analyzed and it was found that the average pregnancy length of the 114 women was 260 days. From a large body of research, it is known that length of human pregnancy has a standard deviation of 16 days.

Based on this study, find a 95% confidence interval for μ , the mean pregnancy length of women who smoke during their pregnancy, and interpret your interval in context

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss E

Q9: A study was done on pregnant women who smoked during their pregnancies. In particular, the researchers wanted to study the effect that smoking has on pregnancy length. A sample of 114 pregnant women who were smokers participated in the study and were followed until the birth of their child. At the end of the study, the collected data were analyzed and it was found that the average pregnancy length of the 114 women was 260 days. From a large body of research, it is known that length of human pregnancy has a standard deviation of 16 days.

Based on this study, find a 95% confidence interval for μ , the mean pregnancy length of women who smoke during their pregnancy, and interpret your interval in context

The problem provides the following information:

- Our sample size is $n = 114$
- This sample gives a sample mean of $\bar{x} = 260$
- Human pregnancy is known to have a standard deviation of $\sigma = 16$.

The 95% confidence interval for μ is therefore:

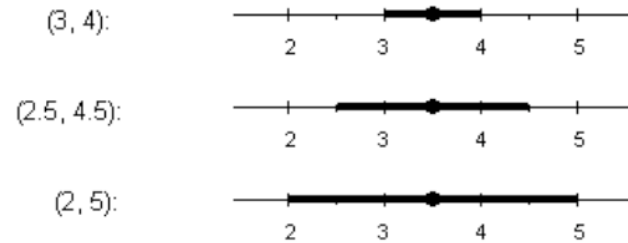
$$260 \pm 2 * \frac{16}{\sqrt{114}} = 260 \pm 3$$

or (257, 263).

Interpretation: We are 95% confident that the mean pregnancy length of women who smoke during pregnancy is covered by the interval (257, 263).

Reflect and Discuss (F)

In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. Based on the study results, the researchers estimated the mean time that a male 25 to 35 years of age spends exercising with 90%, 95%, and 99% confidence intervals. These were (not necessarily in the same order):



Q10: ... the **most** confidence ...

Q11: ... the **most** confidence ...

Q13: ... the **most precise estimation** ...

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q10:** For which of the three intervals do you have the most confidence that it captures the population mean (μ)?

A. (3, 4).

B. (2.5, 4.5).

C. (2, 5).

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q10:** For which of the three intervals do you have the most confidence that it captures the population mean (μ)?

- A. (3, 4).
- B. (2.5, 4.5).
- C. (2, 5).

Answer C

The widest interval gives us the most confidence of capturing the population mean, because it covers more of the number line.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q11:** The confidence interval in which you have the most confidence that it captures the population mean μ must be:

- A. the 90% confidence interval.
- B. the 95% confidence interval.
- C. the 99% confidence interval

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q11:** The confidence interval in which you have the most confidence that it captures the population mean μ must be:

- A. the 90% confidence interval.
- B. the 95% confidence interval.
- C. the 99% confidence interval

Answer C

The confidence percentage measures our confidence that the interval captures the population mean, so the widest interval must be the one with the largest confidence.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q12:** Which of the three confidence intervals provides the most precise estimation?

A. $(3, 4)$.

B. $(2, 5)$.

C. $(2.5, 4.5)$.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss F

 **Q12:** Which of the three confidence intervals provides the most precise estimation?

- A. $(3, 4)$.
- B. $(2, 5)$.
- C. $(2.5, 4.5)$.

Answer A

Since this is the narrowest confidence interval, it is the one that provides the most precise estimation of the unknown mean.

Reflect and Discuss (G-K)

- **Q13:** What is the **margin of error** of this confidence interval?
- **Q14:** ... confidence interval with a smaller **margin of error** ...
- **Q15:** ... a **confidence interval** for μ ...
- **Q16:** -
- **Q17:** The average time full-time corporate employees work per week is **more than 40** hours.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss G

 **Q13:** Suppose that based on a random sample, a 95% confidence interval for the mean hours slept (per day) among graduate students was found to be (6.5, 6.9). What is the margin of error of this confidence interval?

A. 6.7.

B. 0.4.

C. 0.2.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss G

 **Q13:** Suppose that based on a random sample, a 95% confidence interval for the mean hours slept (per day) among graduate students was found to be (6.5, 6.9). What is the margin of error of this confidence interval?

A. 6.7.

B. 0.4.

C. 0.2.

Answer C

Because the width of the confidence interval is 0.4 ($6.9 - 6.5$), the margin of error must be $0.4 / 2 = 0.2$.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss H

➔ Q14: A medical researcher wanted to estimate μ , the mean weight of newborns born to women over the age of 40. The researcher chose a random sample of 100 pregnant women who were over 40, followed them through the pregnancy, and found that the mean weight of the 100 newborns was 3,035 grams. From past research, it is assumed that the weight of newborns has a standard deviation of $\sigma = 500$. The researcher calculated the 95% confidence interval for μ to be (2935, 3135).

If the researcher wanted to maintain the 95% level of confidence but report a confidence interval with a smaller margin of error, which of the following will achieve that?

- A. Redo the study, but this time with a sample of size 64.
- B. Redo the study, but this time with a sample of size 225.
- C. Redo the study and choose a different sample of size 100

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss H

➔ Q14: A medical researcher wanted to estimate μ , the mean weight of newborns born to women over the age of 40. The researcher chose a random sample of 100 pregnant women who were over 40, followed them through the pregnancy, and found that the mean weight of the 100 newborns was 3,035 grams. From past research, it is assumed that the weight of newborns has a standard deviation of $\sigma = 500$. The researcher calculated the 95% confidence interval for μ to be (2935, 3135).

If the researcher wanted to maintain the 95% level of confidence but report a confidence interval with a smaller margin of error, which of the following will achieve that?

- A. Redo the study, but this time with a sample of size 64.
- B. Redo the study, but this time with a sample of size 225.
- C. Redo the study and choose a different sample of size 100

Answer B

A larger sample size reduces the margin of error (for a given level of confidence). In particular, by using a sample size of 225 instead of 100, the margin of error will be reduced from 100 to roughly 67.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss I

 Q15: Below are four different situations in which a confidence interval for μ is called for.

Situation A: In order to estimate μ , the mean annual salary of high-school teachers in a certain state, a random sample of 150 teachers was chosen and their average salary was found to be \$38,450. From past experience, it is known that teachers' salaries have a standard deviation of \$5,000.

Situation B: A medical researcher wanted to estimate μ , the mean recovery time from open-heart surgery for males between the ages of 50 and 60. The researcher followed the next 15 male patients in this age group who underwent open-heart surgery in his medical institute through their recovery period. (Comment: Even though the sample was not strictly random, there is no reason to believe that the sample of “the next 15 patients” introduces any bias, so it is as good as a random sample). The mean recovery time of the 15 patients was 26 days. From the large body of research that was done in this area, it is assumed that recovery times from open-heart surgery have a standard deviation of 3 days.

Situation C: In order to estimate μ , the mean score on the quantitative reasoning part of the GRE (Graduate Record Examination) of all MBA students, a random sample of 1,200 MBA students was chosen, and their scores were recorded. The sample mean was found to be 590. It is known that the quantitative reasoning scores on the GRE vary normally with a standard deviation of 150.

Situation D: A psychologist wanted to estimate μ , the mean time it takes 6-year-old children diagnosed with Down's Syndrome to complete a certain cognitive task. A random sample of 12 children was chosen and their times were recorded. The average time it took the 12 children to complete the task was 7.5 minutes. From past experience with similar tasks, the time is known to vary normally with a standard deviation of 1.3 minutes. In which situation can we not use the confidence interval that we developed

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss I

Q15: Below are four different situations in which a confidence interval for μ is called for.

Situation A: In order to estimate μ , the mean annual salary of high-school teachers in a certain state, a random sample of 150 teachers was chosen and their average salary was found to be \$38,450. From past experience, it is known that teachers' salaries have a standard deviation of \$5,000.

Situation B: A medical researcher wanted to estimate μ , the mean recovery time from open-heart surgery for males between the ages of 50 and 60. The researcher followed the next 15 male patients in this age group who underwent open-heart surgery in his medical institute through their recovery period. (Comment: Even though the sample was not strictly random, there is no reason to believe that the sample of “the next 15 patients” introduces any bias, so it is as good as a random sample). The mean recovery time of the 15 patients was 26 days. From the large body of research that was done in this area, it is assumed that recovery times from open-heart surgery have a standard deviation of 3 days.

Situation C: In order to estimate μ , the mean score on the quantitative reasoning part of the GRE (Graduate Record Examination) of all students who took the exam, a random sample of 100 students was chosen. The mean score was 150. From past experience, it is known that the scores on the quantitative reasoning part of the GRE have a standard deviation of 10.

Answer B

Even though the sample was selected in a manner that is not strictly random, no bias is expected. However, the sample size is small enough (less than 30) to require that the recovery times are normally distributed. Since we do not know this, we cannot use this confidence interval.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss J

→ Q16: In order to estimate μ , the mean weight of a typical passenger + his/her luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers who took the flight. From past research on similar flights, it is known that weights of passengers + their luggage have a normal distribution with standard deviation $\sigma = 20$ pounds.

Can we safely estimate μ with the formula for confidence interval that we developed?

A. Yes

B. No

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss J

Q16: In order to estimate μ , the mean weight of a typical passenger + his/her luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers who took the flight. From past research on similar flights, it is known that weights of passengers + their luggage have a normal distribution with standard deviation $\sigma = 20$ pounds.

Can we safely estimate μ with the formula for confidence interval that we developed?

A. Yes B. No

Answer A

Here is the problem with the important information emphasized. In order to estimate μ , the mean weight of a typical passenger + his/her luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers who took the flight. From past research on similar flights, it is known that weights of passengers + their luggage have a normal distribution with standard deviation $\sigma = 20$ pounds. In the following table, which situation most applies to this situation?

	Small sample size	Large sample size
Variable varies normally	✓	✓
Variable doesn't vary normally	✗	✓

Even though the sample size is only 16, we can safely use the confidence interval for the mean since we are given that the weights have a normal distribution.

	Small sample size	Large sample size
Variable varies normally	✓	✓
Variable doesn't vary normally	✗	✓

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss K

→ Q16: For many years “working full-time” has meant 40 hours per week. Nowadays it seems that corporate employers expect their employees to work more than this amount. A researcher decides to investigate this hypothesis.

- Claim 1: The average time full-time corporate employees work per week is 40 hours.
- Claim 2: The average time full-time corporate employees work per week is more than 40 hours.

To substantiate his claim, the researcher randomly selects 250 corporate employees and finds that they work an average of 47 hours per week with a standard deviation of 3.2 hours. In order to assess the evidence, we need to ask:

- A. how likely it is in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47.
- B. how likely it is that the true mean number of hours per week corporate employees work is 40.
- C. how likely it is that the true mean number of hours per week corporate employees work is more than 40.
- D. how likely it is that in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47 if the true mean is 4

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss K

Q16: For many years “working full-time” has meant 40 hours per week. Nowadays it seems that corporate employers expect their employees to work more than this amount. A researcher decides to investigate this hypothesis.

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- B. how likely it is that the true mean number of hours per week corporate employees work is 40.
- C. how likely it is that the true mean number of hours per week corporate employees work is more than 40.
- D. how likely it is that in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47 if the true mean is 4

Answer D.

Recall that in hypothesis testing, in order to assess the evidence, we need to find how likely is it to get data like those observed assuming that claim 1 is true. In this case, the observed data result in the calculation that the mean number of hours worked is 47 in our sample of 250. Claim 1 is that the true mean is 40. Given our observed data, along with claim 1, what we want to find here is how likely it is that in a sample of 250 we'll find that the mean number of hours worked is as high as 47 (our observed data), when the true mean is actually 40 (claim 1). In hypothesis testing, in order to assess the evidence, we need to find how likely is it to get data like those observed assuming that claim 1 is true.

Reflect and Discuss (L-O)

- **Q18:** What is the **null hypothesis** in this case?
- **Q19:** What is the **alternative hypothesis** in this case?
- **Q20:** The fact that the **$p\text{-value} = 0.002$** means that: ...
- **Q21:** Based on the **$p\text{-value}$** you can conclude that: ...
- **Q22:** Say that the **$p\text{-value}$** was not given, ...
- **Q23:** When would you conclude that the data provide enough evidence that the proportion of Pakistani adults who oppose coeducation is **50%**?
- **Q24:** -
- **Q25:** what could researchers do to reduce the **risk of this error**?

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

➔ Q18: According to the Centers for Disease Control and Prevention, the proportion of U.S. adults age 25 or older who smoke is 0.22. A researcher suspects that the rate is lower among U.S. adults 25 or older who have a bachelor's degree or higher education level. What is the null hypothesis in this case?

- A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22.
- B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
- C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
- D. There is no relationship between education level and smoking habits

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

→ Q18: According to the Centers for Disease Control and Prevention, the proportion of U.S. adults age 25 or older who smoke is 0.22. A researcher suspects that the rate is lower among U.S. adults 25 or older who have a bachelor's degree or higher education level. What is the null hypothesis in this case?

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- B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
- C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
- D. There is no relationship between education level and smoking habits

Answer B

The null hypothesis suggests nothing special is going on; there is no change from the status quo; no difference from the traditional state of affairs. In the context of this problem, the null hypothesis claims that “there is nothing special” about the smoking habits of U.S. adults age 25 and older who have a bachelor's degree. The null hypothesis claims that “nothing special is going on” or “there is no change from the status quo,” which, in this context, means that there is nothing special about the smoking habits of adults age 25 and older who have a bachelor's degree or higher. They are the same as in the entire population of adults (25+).

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

 Q19: What is the alternative hypothesis in this case?

- A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22.
- B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
- C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
- D. There is a relationship between level of education level and smoking habits.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

 Q19: What is the alternative hypothesis in this case?

- A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22.
- B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
- C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
- D. There is a relationship between level of education level and smoking habits.

Answer A.

The alternative hypothesis usually represents what we want to check, or what we suspect is really going on. In this case, what the researcher suspects that is going on is that the proportion of smokers among U.S. adults age 25 or older who have a bachelor's degree or higher is lower than the rate of all adults in this age group. The alternative hypothesis usually represents what we want to check, or what we suspect is really going on. In this case, what the researcher suspects is going on is that the proportion of smokers among U.S. adults age 25 and older who have a bachelor's degree or higher is lower than the rate of all adults in this age group.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

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Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss L

 Q19: What is the alternative hypothesis in this case?

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- B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22.
- C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22.
- D. There is a relationship between level of education level and smoking habits.

Answer – A

The alternative hypothesis usually represent what we want to check, or what we suspect is really going on. In this case, what the researcher suspects that is going on is that the proportion of smokers among U.S. adults age 25 or older who have a bachelor's degree or higher is lower than the rate of all adults in this age group. The alternative hypothesis usually represents what we want to check, or what we suspect is really going on. In this case, what the researcher suspects is going on is that the proportion of smokers among U.S. adults age 25 and older who have a bachelor's degree or higher is lower than the rate of all adults in this age group.

Reflect & Discuss (M)

The following two hypotheses are tested:

- H_0 : The proportion of Pakistani adults who oppose coeducation is roughly 50%.
- H_a : The proportion of Pakistani adults who oppose coeducation is above 50% (i.e., the majority oppose).

Suppose a survey was conducted in which a random sample of 1,100 Pakistani adults was asked about their opinions about coeducation, and based on the data, the p -value was found to be 0.002.

Comment: Throughout this question use a 0.05 (5%) significance level (cutoff).

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Reflect and Discuss M

➡ Q20: The fact that the p -value = 0.002 means that:

- A. There is 0.002 probability of observing data like those observed.
- B. There is 0.002 probability that 50% of Pakistani adults oppose coeducation.
- C. There is a probability of 0.002 (i.e., very unlikely) to observe data like those observed if the proportion of Pakistani adults who oppose coeducation were 50%.
- D. There is 0.998 probability that the majority of Pakistani adults oppose coeducation.

Reading Exercise – No 27

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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Answer C.

The p -value is the probability of observing data like those observed assuming that the null hypothesis, H_0 , is true.

In this case, the p -value is the probability of obtaining the results we did from our sample (specifics of the survey results were not given) if the null hypothesis, H_0 , were true (in other words, if it were true that the proportion of Pakistani adults who oppose coeducation is 50%)

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